Stage 1 & Stage 2 Hard Drill

Chapter 2.3 — Order Statistics & Non-Parametric Extremes

10 new competition-grade questions, full-chapter coverage. Q1,3,5,7,9 target Stage 1 speed (under 90s); Q2,4,6,8,10 reward Stage 2 depth.

Q1 — Extremes Beyond Closed-Form CDFs

$n = 10$ i.i.d. daily log-returns $\sim \mathcal{N}(0,1)$. Using $\Phi(1.28) \approx 0.90$, find $\mathbb{P}(\max \le 1.28)$.
Hint direction: the formula $\mathbb{P}(\max \le x) = [F(x)]^n$ never required F to be invertible or closed-form — it only needs $F(x)$ as a number. Don't let the Normal's lack of a closed-form CDF stop you.

Q2 — Rescaling the Support

$X_1, \dots, X_6$ i.i.d. Unif$(0,2)$ (*not* $(0,1)$). Find the density of $\min(X_1, \dots, X_6)$ and $\mathbb{E}[\min]$.
Hint direction: write $F(x) = x/2$ on $[0,2]$ and substitute directly into the general $f_{\min}$ formula — resist the urge to rescale to Unif$(0,1)$ first and rescale back; doing it directly is faster and less error-prone here.

Q3 — Variance of the Range

$n = 7$ i.i.d. Unif$(0,1)$. The mean range is known: $\mathbb{E}[R] = \frac{n-1}{n+1}$. Find $\mathrm{Var}(R)$.
Hint direction: $R \sim \mathrm{Beta}(n-1, 2)$ — use the general Beta variance formula $\frac{ab}{(a+b)^2(a+b+1)}$ with $a = n-1, b = 2$, don't re-derive it from the density.

Q4 — A Deliberate Trap: Does the Speed Trick Still Work?

$X_1, \dots, X_5$ i.i.d. Exp(1). Find $\mathbb{P}(\min > 1,\ \max < 3)$.
Hint direction: before reaching for the joint-density formula $f_{(1),(n)}(a,b) = n(n-1)[F(b) - F(a)]^{n-2} f(a) f(b)$, ask whether Example 5's "all n values land in one interval" shortcut required F to be uniform, or just continuous. Re-read your own formula sheet's speed-trick note carefully.

Q5 — Naming the Order Statistic

$n = 11$ i.i.d. Unif$(0,1)$. State the exact name and parameters of the distribution of $X_{(3)}$, and give $\mathbb{E}[X_{(3)}]$ with no integration.
Hint direction: two one-line facts chain together here — the general $X_{(k)} \sim \mathrm{Beta}(k, n-k+1)$ result, and the spacing-sum mean shortcut.

Q6 — Capstone: Revenue Equivalence

$n = 7$ risk-neutral bidders, valuations i.i.d. Unif$(0,1)$. In the symmetric first-price sealed-bid equilibrium, each bidder shades their bid to $b(v) = \frac{n-1}{n} v$. Find the seller's expected revenue under first-price rules, and compare it to the expected revenue under second-price (Vickrey) rules.
Hint direction: expected first-price revenue is $\mathbb{E}[b(X_{(n)})] = \frac{n-1}{n}\mathbb{E}[X_{(n)}]$; expected second-price revenue is $\mathbb{E}[X_{(n-1)}]$ (Example 9's logic) — compute both from the $k/(n+1)$ formula and see what you discover. This "coincidence" has a name.

Q7 — Spacings Arithmetic, Fast

$n = 14$ points break $[0,1]$ into 15 spacings. State $\mathbb{E}[S_i]$ for any i, and find $\mathrm{Var}(X_{(7)})$ (the near-median order statistic).
Hint direction: both answers are a single formula substitution away — this is a "how fast can you plug in" check, not a derivation.

Q8 — Covariance Between Two Order Statistics

For n i.i.d. Unif$(0,1)$ and $j < k$, derive $\mathrm{Cov}(X_{(j)}, X_{(k)})$ in terms of j, k, n. Then evaluate it for $n = 9, j = 2, k = 6$.
Hint direction: write $X_{(j)} = S_1 + \cdots + S_j$ and $X_{(k)} = S_1 + \cdots + S_k$ (shared spacings $S_1, \dots, S_j$ drive the covariance), and reuse Example 14's variance formula as the $j = k$ sanity check on whatever general formula you derive.

Q9 — Rényi Representation Meets the Beta Marginal

$n = 3$ points break $[0,1]$ into 4 spacings. Using the fact that every individual spacing $S_i \sim \mathrm{Beta}(1, n)$ (Example 1-type reasoning, not the joint representation), find $\mathbb{P}(S_1 > 0.5)$.
Hint direction: Beta$(1,n)$ has CDF $1 - (1-x)^n$ — this is a one-line computation once you recognize which named distribution S_1 already is.

Q10 — Capstone: Order-Book Gap Risk

$n = 9$ resting limit orders land at i.i.d. Unif$(0,1)$ normalized price levels, creating 10 spacings (gaps) between consecutive price levels (including the two boundary gaps). Using the general "all spacings $\le c$" inclusion–exclusion formula, find $\mathbb{P}(\text{every gap} \le 0.3)$.
Hint direction:

$$\mathbb{P}\left(\max_i S_i \le c\right) = \sum_{j=0}^{\lfloor 1/c \rfloor} (-1)^j \binom{n+1}{j} (1 - jc)_+^n$$

with $n+1 = 10$ spacings total here — be careful that the formula's "n" is the *number of spacings minus one*, i.e. the number of break points, not the number of spacings itself. Work out how many terms $j = 0, 1, 2, \dots$ actually survive before $(1-jc)_+$ hits zero.

No solutions above by design — Q1,3,5,7,9 under a 90-second timer (Stage 1 calibration); Q2,4,6,8 closed-book with full derivations (Stage 2 calibration); Q10 as a longer sit-down problem. Report which ones took longer than expected — that gap is your next drill target.